

Calibrated DistilBERT for Reliable Extractive QA

Project milestone | Default project, PCE division | Independent course study

Abstract

We investigate reliable abstention for extractive question answering on SQuAD 2.0. A compact DistilBERT checkpoint trained only on SQuAD 1.1 provides strong span extraction but no useful default behavior on unanswerable questions. We implemented a complete SQuAD 2.0 pipeline, fine-tuned the model on 12,000 examples, and calibrated the no-answer threshold on 1,500 held-out examples. On a separate 1,500-example evaluation set, fine-tuning improved F1 from 41.79 to 63.92; calibration further increased it to 64.67. The final system obtains 62.53 EM, 64.67 F1, and 70.94% unanswerable accuracy.

Approach

The model jointly encodes each question and context using a six-layer DistilBERT transformer. A linear QA head assigns start and end logits to every token. For each context window we enumerate valid spans from the top 20 start and end positions, retain the highest-scoring span up to 30 tokens, and compare it with the [CLS] no-answer score.

Long passages use 256-token windows with a 64-token stride. During training, answer spans outside a window and genuinely unanswerable examples both target [CLS]. During inference, the score difference $d = s(\text{null}) - s(\text{best span})$ controls abstention: the model emits an empty answer when d exceeds threshold τ .

Implementation status

- Data loading, token alignment, overflow windows, training, mixed precision, checkpointing, and plotting are complete.
- Metric normalization and F1/EM logic are covered by unit tests; a 32-example smoke run passed end to end.
- The full two-epoch GPU experiment and held-out evaluation are complete.

Experiments and preliminary results

Seeded SQuAD 2.0 subset; final evaluation kept separate from calibration

Experimental setup

We sampled 12,000 training examples with seed 224, producing 13,322 overlapping features. A shuffled 3,000-example validation subset was split evenly: 1,500 examples for threshold selection and 1,500 for final evaluation. The latter contains 743 answerable and 757 unanswerable questions. We trained for two epochs using AdamW, learning rate $3e-5$, weight decay 0.01, effective batch size 24, 10% warmup, gradient clipping at 1.0, and bfloat16 on an RTX 5060 Laptop GPU.

System	EM	F1	Ans. F1	No-answer acc.
SQuAD 1.1 checkpoint	38.00	41.79	84.37	0.00
Fine-tuned on SQuAD 2.0	61.27	63.92	67.53	60.37
Fine-tuned + calibration	62.53	64.67	58.29	70.94

Preliminary interpretation

The SQuAD 1.1 checkpoint answers every unanswerable evaluation question at the default threshold, so its unanswerable accuracy is 0%. Fine-tuning creates a meaningful no-answer signal: without calibration, unanswerable accuracy rises to 60.37% and overall F1 rises by 22.12 points. Threshold calibration moves τ to -1.828, raising unanswerable accuracy to 70.94% and overall F1 by a further 0.76 points.

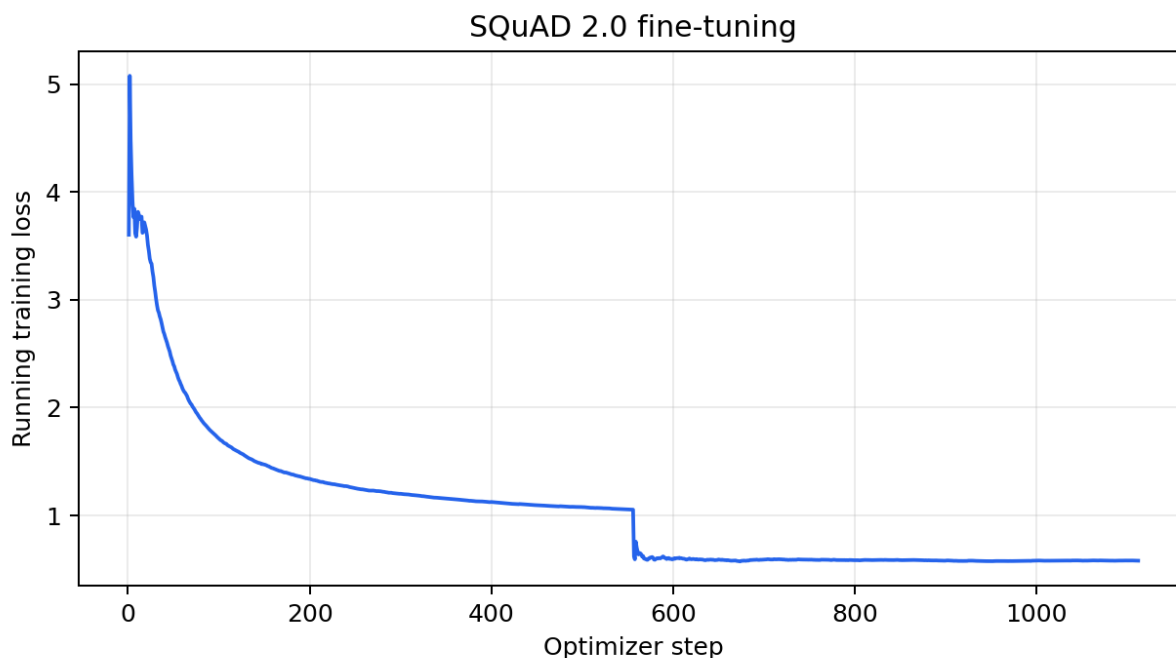


Figure 1. Running loss across 1,111 optimizer steps; the apparent epoch reset reflects a per-epoch running mean.

Progress, analysis, and remaining work

What has been learned and what the final report will emphasize

Qualitative progress

Fine-tuning corrects both answer extraction and abstention. For the question 'What type of behavior in primes is it possible to determine?', the baseline abstains while the adapted model returns 'statistical', matching a reference. For the adversarial question 'What was the Yuan's Persian enemy?', the context mentions an ally in Persia but no enemy; the baseline chooses 'the Ilkhanate', whereas the adapted system correctly abstains.

Regressions remain. The adapted model sometimes selects a semantically complete but annotation-mismatched boundary, such as 'About 61.1%' instead of '61.1%'. It also becomes overconfident in abstention on some answerable examples. These cases motivate reporting token F1 beside exact match and separating answerable from unanswerable results.

Progress against plan

- Completed: repository recovery, modern pipeline, tests, smoke run, full training, calibration, independent evaluation, model checkpoint, plots, and error records.
- Completed: quantitative comparison of three operating points and breakdown by question answerability.
- In progress at milestone: converting the evidence into the final report and poster.

Final-report plan

The final report will clearly distinguish architecture transfer, task adaptation, and calibration. It will include the aggregate table, training curve, selected improvements and regressions, limitations of subset evaluation, compute cost, and a reproducibility checklist. No leaderboard claim will be made because the experiment uses a seeded subset of the public validation data rather than the course's hidden test split.

Current conclusion

The strongest result is not merely the 22.88-point F1 increase from task adaptation; it is the change in error balance. Calibration trades some answerable recall for much better abstention, and the disaggregated metrics make that trade-off visible. This is the central lesson to carry into the final analysis.